

Asymmetric Gradient Boosting for Carbon Forecasting and Deadline-Safe Green-Window Scheduling

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Abstract. Standard grid-computing baselines rarely couple environmental forecasts with resource allocation: first-come-first-served (FCFS) scheduling disregards carbon signals, while direct carbon-footprint (CFP) regression ignores the physical link between energy and emissions. We present *GreenGuardian*, a two-stage system for the GreenDIGIT ECML-PKDD 2026 Challenge: an asymmetric LightGBM forecaster trained on a pseudo-Huber surrogate objective and selected by validation symmetric mean absolute percentage error (sMAPE), paired with physical carbon interpolation, and a *GreenWindowScheduler* combining a multi-resource capacity ledger with a slack-bounded Deadline Guard. On Task A, our forecaster improves ScoreTA from 0.765 to **0.644**. On Task B (Nov. 2025–Mar. 2026, 28,662 jobs) against the carbon-blind FCFS reference, using the challenge’s offline signal table, *GreenGuardian* reduces carbon by **89.9%** and energy by **89.1%**, with unchanged makespan, **100.0%** deadline compliance, and a final score of **0.732**.

Keywords: Carbon-Aware Scheduling · Energy Forecasting · Gradient Boosting · Green Computing · Grid Infrastructures.

1 Background

Digital research infrastructures contribute a growing share of global greenhouse gas emissions. The GreenDIGIT project [6] addresses this through data-driven accounting, centered on the Environmental Impact Metric Publication System (EIMPS), which unifies sustainability metrics from heterogeneous Grid/Cloud/Network environments into standardized time series. The ECML-PKDD 2026 Challenge builds on EIMPS data: forecasting site-level energy and carbon footprint (CFP, Task A) to drive sustainability-aware scheduling (Task B).

Competition goals. The tasks form a forecast-then-act pipeline. **Task A:** for each site and 15-minute origin, predict total energy (**energy_wh**) and carbon footprint (**cfp_g**) at 1-hour ($h=4$ steps) and 24-hour ($h=96$ steps) horizons, plus two per-bucket classification subtasks—A.1 flags missing or invalid signals, A.2 flags peak (anomalously high-usage) events. **Task B:** a discrete-event simulator replays a

real job trace, and the scheduler decides *when* and *where* each job runs from a per-bucket energy and carbon signal table, minimizing energy and carbon while meeting deadlines, measured against a carbon-blind First-Come-First-Served (FCFS) reference. The stages are linked in principle: a scheduler can shift load only as far as the signals driving it are reliable.

Data. Task A aggregates site metrics into 15-minute buckets (`bucket_15m`): sample counts (`records`), `energy_wh` (Wh), and `cfp_g` (gCO₂eq); the raw feed adds aggregate compute `work` and core counts (`ncores`), used for per-core normalization. Three production sites span ~ 3.7 months each at 85–90% continuity. The official split trains on buckets at or before 2026-02-18T14:00:00Z and tests on later ones, so any feature exposing a future value constitutes leakage.

Baseline weaknesses. The Task A reference regresses `cfp_g` independently of `energy_wh`, ignoring their physical coupling, without leakage control or variance reduction. On Task B, FCFS ignores carbon entirely, while the `greedy_carbon` baseline checks only instantaneous slot occupancy under a static hold threshold, so independently deferred jobs can converge on and exhaust the same low-carbon interval. *GreenGuardian* addresses both through leakage-controlled forecast features and deterministic capacity accounting.

2 Methodology

2.1 Task A: Asymmetric Forecasting and Physical Carbon Interpolation

We train direct (non-recursive) LightGBM regressors [1]—one per target and horizon, four models—to avoid compounding recursive error. Features combine cyclical time encodings, causal lags ($\{1, 4, 96, 2, 8, 16, 192, 672\}$ steps, up to 7 days), rolling moments on *shifted* series (3h mean/std, 24h mean/std/ q_{95}), per-core normalized signals (`energy_per_core`, `cfp_per_core`, `work_per_core`), and momentum differences ($\text{lag}_1 - \text{lag}_4$, $\text{lag}_4 - \text{lag}_{96}$); no bucket sees its own future value. Squared error is symmetric, whereas under- and over-forecasting carry asymmetric downstream costs, so we train on a surrogate rather than on sMAPE itself. Let $e = \hat{y} - y$ in $\log(1 + x)$ space, $u = e/\delta$ ($\delta = 0.18$), and $a(e) = 1.06$ if $e < 0$, else 0.94. The surrogate’s gradient and Hessian are

$$g(e) = a(e) \frac{e}{\sqrt{1 + u^2}}, \quad h(e) = \max\left(a(e) (1 + u^2)^{-3/2}, 0.08\right), \quad (1)$$

a smooth pseudo-Huber approximation to Huber’s robust loss [5] biased against under-prediction. The surrogate is not mathematically consistent with sMAPE; model selection evaluates true sMAPE in original space on a held-out validation window. A 3-seed bagged ensemble reduces variance; cold-start series fall back to a seasonal (day-of-week/hour/minute) median. Rather than trusting an independent `cfp_g` regressor, we exploit the energy-carbon coupling: for site

s , $R_s = \text{median}_{t \in \text{train}}(\text{cfp}_{s,t}/\text{energy}_{s,t}) \in [0, 10] \text{ gCO}_2\text{eq/Wh}$ is a training-only emission ratio, and the final forecast is

$$\widehat{\text{cfp}}_{s,t} = \lambda \widehat{\text{cfp}}_{s,t}^{\text{ML}} + (1 - \lambda) R_s \widehat{\text{energy}}_{s,t}, \quad (2)$$

with $\lambda = 0.18$ (1h) and $\lambda = 0.12$ (24h), shrinking toward the physical estimate as horizon-induced variance grows.

2.2 Task A: Detection and Peak Thresholding

Signal-validity scores blend site (ρ_s), quarter-hour-of-day (ρ_q , 96 daily 15-minute slots), and global ($\bar{\rho}$) rates as $p_{\text{valid}} = (\rho_s + 8\rho_q + \bar{\rho})/10$, reflecting an empirical prior that missingness is predominantly time-correlated. Peak events use a per-site 95th-percentile threshold θ_s and anomaly score $\text{clip}(r/(1+r), 0, 1)$ with $r = \max(\widehat{\text{energy}}, \widehat{\text{cfp}})/\theta_s$, computed from our own Task A predictions. Two thresholds govern deployment: the F_1 -optimal $\tau_{F_1} = \arg \max_{\tau} F_1(\tau)$ on training data, and the deployed $\tau^* = \max(0.01, \min(\tau_{F_1} - 0.05, 0.20))$, shifted toward recall to reduce false negatives.

2.3 Task B: GreenWindowScheduler

Each tick, the scheduler receives the pending queue (deadlines, priorities, site whitelists), a registry of site slot/core/memory capacities, and a per-bucket energy and carbon signal table; it returns a site and dispatch time per ready job. The core idea is to hold a job until a low-carbon window [2, 3] while preserving feasibility and deadlines. Jobs are processed earliest-deadline-first (EDF) [4], EXPRESS priority tie-broken forward, with hold horizon $H_j = \min(H_{\text{max}}, 0.95(d_j - t))$, $H_{\text{max}} = 72 \text{ h}$. In the evaluation WMS, d_j is the latest allowed *dispatch start*, so $(d_j - t)$ is exactly the allowable hold window; where deadlines bound *completion*, the guard must instead subtract expected runtime, $H_j = \min(H_{\text{max}}, d_j - \hat{r}_j - \epsilon)$. Low-slack and EXPRESS jobs bypass the search and dispatch immediately at their whitelisted site. Unlike `greedy_carbon`'s instantaneous check, a forward reservation ledger over 15-minute buckets tracks slots, cores, and memory: admission to every bucket b a job would span at site s requires $|J_{b,s}| + 1 \leq S_{\text{max}}^s$, $\sum_{j' \in b,s} c_{j'} + c_j \leq C_{\text{max}}^s$, and $\sum_{j' \in b,s} m_{j'} + m_j \leq M_{\text{max}}^s$ before a hold is committed, preventing deferred jobs from exhausting the same low-carbon bucket.

Candidate windows are evaluated against a cached signal table; in the offline evaluation this is the challenge's per-bucket table, pre-loaded at initialization, and ground-truth intensities are never queried at dispatch time. A window is accepted only if the relative carbon gain $\Delta_c = (c_{\text{now}} - c_{\text{future}})/c_{\text{now}}$ exceeds $\theta_0 + \min(\theta_{\text{cap}}, k \cdot H_{\text{hold}})$ with $\theta_0 = \theta_{\text{cap}} = 0.01$, $k = 0.001 \text{ h}^{-1}$ (1% minimum, capped at 2%), while the relative energy penalty $\Delta_e = (e_{\text{future}} - e_{\text{now}})/e_{\text{now}}$ stays below 10%. Candidates are ranked after per-site mean normalization of both signals, $\tilde{c} = c/\bar{c}_s$ and $\tilde{e} = e/\bar{e}_s$, by the weighted sum $w_c \tilde{c} + w_e \tilde{e}$, with $(w_c, w_e) = (0.80, 0.20)$ under the carbon objective and $(0.60, 0.40)$ under an energy-leaning declaration; the latter has no completion-time term—deadlines

are protected by the slack guard, not the ranking—so we do not call it a makespan objective. In live API mode the cache instead holds only Task A bundles issued at or before the tick, so the search terminates at $t+24$ h and $H_{\max}$ is a nominal ceiling; the headline results below use the offline table for reproducibility.

3 Results

Evaluation metrics. Forecast accuracy uses $\text{sMAPE} = \frac{1}{n} \sum_i |\hat{y}_i - y_i| / ((|y_i| + |\hat{y}_i|)/2)$. Per horizon, $S(A)_h = \frac{1}{2} \text{sMAPE}_h^{\text{energy}} + \frac{1}{2} \text{sMAPE}_h^{\text{cfp}}$; for the auxiliary subtasks, $S(A)_k = 1 - \frac{1}{2}(\text{AUROC}_k + F_{1,k})$; and $\text{ScoreTA} = 0.70 \cdot \frac{1}{2}(S(A)_{1h} + S(A)_{24h}) + 0.15 S(A.1) + 0.15 S(A.2)$, for which *lower is better*. For Task B, each objective $x \in \{\text{energy, carbon, makespan}\}$ gives $\Delta_x = (b_x - s_x)/b_x$ (baseline b_x , submission s_x ; positive is better), combined as $\text{Pareto} = 1 - \frac{1}{3} \sum_x (1 - \Delta_x)$ and $\text{Score} = \text{Pareto} + 0.15 \max(0, \Delta_{\text{declared}}) - P_{\text{deadline}}$, for which *higher is better*. The conventions are opposite, so ScoreTA 0.644 versus the baseline’s 0.765 is an improvement, while the Task B 0.732 is on the higher-is-better scale.

Table 1 reports all scores; the four detection components make ScoreTA reconstructable. Forecasting shows the expected sMAPE degradation from 1h to 24h. The detection subtasks differ sharply. On the A.1 test window ($N=5,902$), valid buckets are 99.88% (5,895) against 7 invalid; the deployed detector labels every bucket valid ($\text{TP} = 5,895$, $\text{FP} = 7$, $\text{TN} = \text{FN} = 0$), so invalid-class recall is 0.000 and macro- F_1 and balanced accuracy are both 0.500—chance level. The headline F_1 of 0.999 is a prevalence artifact, consistent with the modest AUROC (0.573). A.2 reaches AUROC 0.797 at F_1 0.321, the price of a recall-oriented threshold. On Task B, makespan is unchanged because EXPRESS and low-slack jobs dictate the batch tail and dispatch immediately, so deferring flexible jobs never extends the schedule; FCFS holds nothing, so its compliance is bounded only by capacity, and the offline run’s 100.0% rate shows deferral caused no feasibility regression.

Attribution. The trace pins every one of the 28,662 jobs to exactly one site (whitelist cardinality 1), so no scheduler can re-route work spatially; all gains are temporal. Under the simulator’s time-indexed accounting, a job is charged the site energy and carbon of its dispatch bucket, scaled by duration, so temporal deferral moves jobs to lower-valued energy and carbon buckets. The simulator does not expose a separate causal decomposition of occupancy and temporal-intensity effects, so the split between the energy channel and the carbon coupling of Eq. 2 cannot be read off the reduction percentages. Two properties of the ranking were checked. First, because both signals are normalized by per-site means before weighting, the reported weights act on comparable scales; substituting raw magnitudes shifts the dispatch time of a subset of jobs but leaves the final score unchanged (0.732), so the result is robust to that choice. The Task A forecaster of Section 2.1 is our forecasting submission; the Task B scheduler is evaluated on the challenge’s provided signal table, so the two results are reported independently rather than as an end-to-end coupled pipeline. Second, the headline run uses

the challenge’s offline signal table, which admits lookups whose forecast origins postdate the tick; we therefore report it as an offline, oracle-style scheduling result rather than a live-forecast result. A one-day live-API smoke run reached 0.500 (0.4997 exact) on 304 jobs; because it differs in period and workload from the full offline run, it is evidence of operational compatibility rather than a controlled estimate of the value of additional forecast reach. We attribute the tolerance to high sMAPE (0.718–0.841) to grid intensity spanning two orders of magnitude (22.34–3,000.00 gCO₂/kWh; mean 386.69, std 505.51), so the hold logic needs only a coarse peak-versus-valley ordering; a direct measurement of that ordering (rank correlation or oracle regret) is left to future work.

Table 1. Final results versus official baselines.

Task A – Forecasting	
ScoreTA – baseline	0.765
ScoreTA – GreenGuardian	0.644
1h sMAPE (Energy / CFP)	0.718 (0.682 / 0.755)
24h sMAPE (Energy / CFP)	0.841 (0.759 / 0.922)
A.1 Signal Quality (AUROC / F_1)	0.573 / 0.999
A.1 prevalence / invalid recall	99.88% / 0.000
A.1 macro- F_1 / balanced acc.	0.500 / 0.500
A.2 Peak Detection (AUROC / F_1)	0.797 / 0.321
Detection scores $S(A.1)$ / $S(A.2)$	0.214 / 0.441
Task B – Scheduling (vs. FCFS, Nov. 2025–Mar. 2026)	
Jobs simulated / dispatched	28,662 / 28,662
Deadline hit rate (offline)	100.0%
Total carbon – FCFS	3,127,038.2 gCO ₂ eq
Total carbon – GreenGuardian	314,521.9 (-89.9%)
Total energy – FCFS	7,279.4 kWh
Total energy – GreenGuardian	791.1 kWh (-89.1%)
Makespan change	+0.0%
Pareto / bonus / penalty	0.597 / 0.135 / 0.000
Final score – offline (headline)	0.732
Live-API smoke run (304 jobs, 1 day)	0.500

4 Conclusions

With every job pinned to a single site, *GreenGuardian*’s reductions are attributable to temporal shifting under the simulator’s time-indexed cost model: deferral moves jobs to lower-valued energy and carbon buckets, and carbon follows through the physical energy-carbon coupling. The headline result is an offline scheduling outcome, since it draws on the challenge’s per-bucket signal table rather than live forecasts; a one-day live-API run confirms the same scheduler operates against a

streaming Task A service. Point accuracy appears to matter less than preserving a coarse low- versus high-carbon ordering—an assumption consistent with our results but not yet directly measured. Measuring that ordering, quantifying the value of additional forecast reach under a matched workload, and validating on traces that permit spatial routing remain future work.

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